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ERP and fMRI correlates of endogenous and exogenous focusing of visual-spatial attention

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Abstract

The aim of this study was to investigate the neural correlates of the functional distinction underlying attentional mechanisms of endogenous-sustained and exogenous-transient spatial selection. We recorded event related potentials (ERPs) and used functional magnetic resonance imaging (fMRI) in separate experiments while subjects performed a simple reaction time (RT) to the same visual stimulus displayed to one of several field locations. Endogenous-sustained or exogenous-transient focusing of attention onto target location were obtained by presenting the stimulus in blocks of same-point vs. randomised-point trials, respectively. Same-point stimuli yielded overall faster RT than randomised stimuli, indicating a facilitating effect of endogenous-sustained spatial attention on the perceptual processing of the impending stimulus. Moreover, same-point vs. randomised presentations revealed significant increases in the fMRI signal in the bilateral lingual and fusiform gyri as well as in the right calcarine sulcus, in conjunction with a larger amplitude of the posterior P1 component of ERPs, but no modulation of the amplitude of the N1 component. Rather, a larger amplitude of N1 was found in the reverse contrast, randomised minus same-point trials, which revealed increases in the fMRI signal along the posterior left superior frontal sulcus and bilaterally in the superior precuneus. These findings indicate that N1 indexes exogenous orienting of attention and is likely to represent the activity of frontal and parietal components of the attention network involved in eliciting attention changes. In contrast, the effects of those changes, resulting in a modulation of activation in visual occipital areas, are indexed by P1.

Introduction

Spatial selective attention plays a key role in human perception by enhancing the efficiency of visual processing through sensory gain control in the visual centres (Hillyard *et al*., 1998). This has been suggested by the evidence of a modulation of the amplitude of the early P1 (latency of 90–130 ms) and N1 (130–200 ms) components of the event related potential (ERP) over posterior scalp sites contralateral to the attended visual field (see Mangun *et al*., 1993; Mangun, 1995; Hillyard & Anllo-Vento, 1998; for reviews). Moreover, dipole modelling of ERP sources in relation to MRI-defined cortical anatomy (Clark & Hillyard, 1996) or ERP recording combined with PET (Heinze *et al*., 1994; Mangun *et al*., 1997; Woldorff *et al*., 1997; Mangun *et al*., 2001) and functional magnetic resonance imaging (fMRI; Mangun *et al*., 1998; Martínez *et al*., 1999; Martínez *et al*., 2001; Noesselt *et al*., 2002) found increased activity in several extrastriate cortical areas associated to the effect on ERP amplitude. Most of these studies have been concerned with endogenous-sustained attention and it is still unclear whether the same neural mechanisms might also be involved in stimulus driven focusing of attention. Furthermore, several reports of dissociation between the P1 and N1 attentional effects have indicated that they might reflect different aspects of early spatial selection (Hillyard *et al*., 1985; Heinze *et al*., 1990; Luck *et al*., 1990; Mangun & Hillyard, 1991; Luck, 1995). According to one proposal, the P1 and N1 effects reflect suppression and enhancement of sensory processing, respectively, occurring as a consequence of attentional orienting (Luck, 1995). In contrast with this possibility there is evidence of a specific relationship between P1 (but not N1) effect and increased activation in extrastriate cortical areas (Heinze *et al*., 1994; Mangun *et al*., 1997; Woldorff *et al*., 1997; Mangun *et al*., 1998; Noesselt *et al*., 2002), where the P1 neural generators have been typically localized (Clark *et al*., 1995; Di Russo *et al*., 2001). Moreover, there have been reports of an association between N1 and switch of attention to a task-relevant spatial location (Luck *et al*., 1990; Yamaguchi *et al*., 1995). Taken together, this evidence suggests that while P1 could index a preset bias and a consequent sustained gating of input at the attended location, N1 might be related to a shift of attention triggered by potentially relevant events.

In the present study, we compared behavioural, ERP and fMRI responses to the same visual stimulus presented under conditions favouring either an endogenous-sustained or an exogenous-transient allocation of attention. We used a speeded simple detection task to avoid a possible confound between the effect due to spatially specific mechanisms of attention and that related to stimulus discrimination, as might have been the case in previous studies addressing similar issues (e.g. Mangun & Hillyard, 1991). We aimed at investigating the neural

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correlates of spatially specific attention modulations of sensory input dependent upon endogenous-sustained vs. exogenous-transient mechanisms of attention selection. In particular, we wanted to explore the functional significance of a dissociation between P1 and N1 effects.

Materials and methods

General procedure

Using an experimental procedure similar to that employed in our previous behavioural studies (Marzi *et al*., 2002; Natale *et al*., 2005), we manipulated the predictability of the location of the impending stimulus in a simple reaction time paradigm. A black and white checkerboard was presented to one of several locations along the horizontal meridian of the visual field under two different modes of stimulus presentation: a blocked-point condition, in which stimuli were presented predictably to one and the same field location in a given block, and a randomised-point condition, in which stimuli were presented unpredictably to one of several possible locations. Our rationale was that predictability of stimulus location would entail a sustained, endogenous, cognitively driven focusing of attention, whereas unpredictability of stimulus location would encourage a diffuse attention to the whole set of possible stimulus locations followed by a transient, exogenous, stimulus driven focusing of attention after stimulus onset. We employed this simple reaction time (RT) paradigm in two separate experiments with different groups of participants. We recorded visual ERPs in the first experiment and in the second we used fMRI techniques with similar stimuli.

ERP experiment

Subjects

Seven healthy subjects (four women) participated in the ERP experiment, each having given prior informed consent. The experiment was carried out according to the principles laid down in the Declaration of Helsinki and was approved by the Ethical Committee of the University of Verona. All subjects were right-handed, as assessed by the Edinburgh Inventory (Oldfield, 1971). Subjects' age ranged between 18 and 26 years, with a mean age of 21.4 years. All subjects had normal or corrected to normal vision.

Stimuli, apparatus and procedure

Subjects were seated in a dimly illuminated room, facing a screen monitor with the head restrained by a chin rest. Stimulus presentation, event timing and response recording were controlled by the Micro Experimental Laboratory software (MEL2; Schneider, 1995). One black and white checkerboard was displayed for 100 ms onto the black background of the screen monitor in one of seven positions located at 0° and at 4°, 8° and 12° of eccentricity in the left and in the right hemifield along the horizontal meridian just above a central fixation cross. Seven white filled circles of 0, 4° of visual angle in radius marked the possible stimulus locations throughout a whole trial. Subjects were asked to keep fixation onto the central cross during each block of trials and to press a button as quickly as possible with the index finger of either the right or the left hand (order counterbalanced across subjects) following stimulus appearance. The subjects' fixation was continuously monitored by means of a video camera and by electrooculographic recording. Each subject was tested in one session including four blocked-point and four randomised conditions alternated by using an ABBABAAB design in a counterbalanced way across subjects. In each blocked-point condition there was one block of 28 trials for each stimulus location, with the succession of positions following a fixed left-to-right or right-to-left sequence. Those sequences were alternated in the four block-point conditions of an experiment by using an ABBA design in a counterbalanced way across subjects. In the randomised condition there were seven blocks of 28 trials, four for each stimulus location. A total of 1568 trials were presented in each experiment (112 repetitions per seven field locations per two conditions). The experimenter indicated at the beginning of each block whether the stimulus would be presented in a randomised or in a blocked order, showing the exact position of stimulus appearance in the latter case. The interstimulus interval varied between 1300 and 1800 ms plus RT, with a maximum time limit of 2 s for response recording.

Electrophysiological recordings

The electroencephalogram (EEG) was recorded from 28 tin electrodes mounted on an elastic cap, at the following sites (10–20 System): Fp1, Fp2, F3, F4, C3, C4, CP1, CP2, CT5, CT6, P3, P4, T3, T4, T5, T6, PO1, PO2, TO1, TO2, O1, O2, IN3, IN4, Cz, Fz, Ipz, INz (see Fig. 1 for electrode position on the scalp). The right-mastoid electrode served as on-line reference. ERP waveforms were re-referenced off-line to the average of the right- and the left-mastoid electrodes. Two electrodes placed in a bipolar montage at approximately 1 cm from the external canthus of both eyes served to record the horizontal electrooculogram (HEOG). The VEOG (vertical EOG) and blinks were recorded from one electrode positioned below the right eye and referenced to the right mastoid. For all channels the EEG and the EOG signals were amplified (gain 1000), filtered (band-pass filter 0.01–100 Hz), and sampled at A/D rate of 500 Hz.

Data analysis

Average waveforms were computed for each stimulus eccentricity in the two attentional conditions in a time window between 200 ms

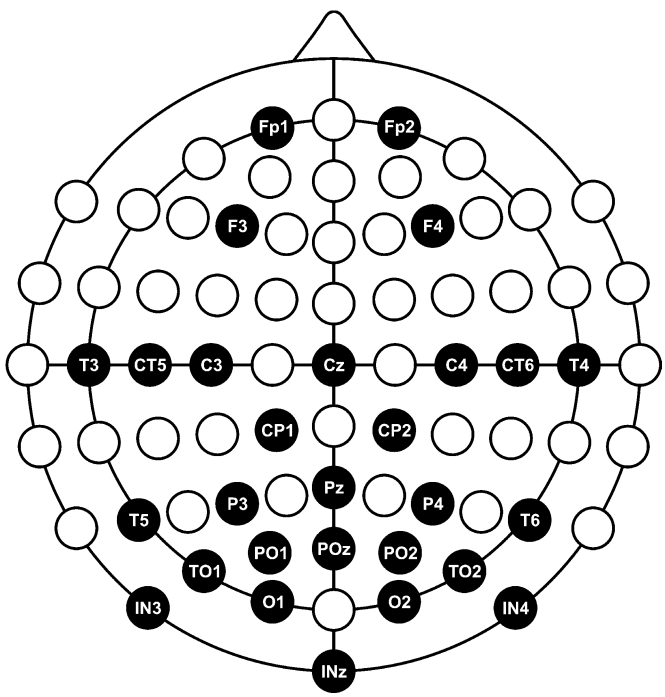

FIG. 1. Map of electrode position on the scalp. According to the new nomenclature (American Electroencephalographic Society, 1994), T5, T6, TO1, TO2, PO1, PO2 are meant to be P7, P8, PO7, PO8, PO3, PO4, respectively.

pre- and 600 ms poststimulus. Trials in which saccades, blinks, and behavioural errors (anticipations, namely RTs faster than 130 ms; retards, namely RTs slower than 800 ms; misses) occurred, were discarded. On average, this led to the rejection of 18.50% of the trials. No difference in the percentage of rejected trials as a function of spatial location or attentional condition was found. The mean amplitude of the P1 and N1 components was measured, in the unfiltered averages, as the mean voltage between 110 and 150 ms poststimulus for P1 and between 155 and 195 ms for N1. The mean amplitudes were calculated relative to the 200-ms prestimulus baseline interval. Three posterior electrode sites in the right (PO2, TO2, T6) and in the left (PO1, TO1, T5) hemisphere were considered for this analysis.

Two five-way ANOVAs, one on P1 and another on N1 mean amplitude values, were carried out with attentional Condition (Blocked-point vs. Randomised), Hemifield (Left vs. Right), Eccentricity (4°, 8°, 12°), Hemisphere (Contra- vs. Ipsilateral to the stimulated hemifield), and Electrode (PO, TO, T) as factors. All the *P*-values reported reflect the Greenhouse-Geisser epsilon correction for nonsphericity (Jennings & Wood, 1976).

Scalp topographies of the attentional modulations of the amplitude of the two components were assessed by performing current source density (CSD; Perrin *et al.*, 1989) maps around the peak of the components and within the time windows considered for mean amplitude measurements and for statistical analysis (110–130 ms for P1 and 165–185 ms for N1).

Mean RT to stimuli presented to each field location in the two conditions was computed for each subject on the same trials used for the ERP averaging procedure. The RT values were analysed with a three-way ANOVA with attentional Condition, Hemifield, and Eccentricity as within-subjects factors. In addition, in order to verify whether possible differences between conditions may persist for subsets of data with similar RT, we have plotted the cumulative frequency distribution of RTs in the blocked and in the randomised condition and run a two-way ANOVA with attentional Condition and Percentile as within-subject factors. Finally, we performed separate analyses on RTs to central stimuli considering both mean values and the cumulative frequency distribution of RTs.

fMRI experiment

Subjects

Eight healthy subjects (three women) participated in the fMRI experiment, each having given prior informed consent according to the Max-Plank-Institute guidelines. All subjects were right-handed, as assessed by the Edinburgh Inventory (Oldfield, 1971). Subjects' age ranged between 21 and 33 years, with a mean age of 25 years. All subjects had normal or corrected to normal vision. The fMRI procedures were approved by the local ethics review board at the University of Leipzig.

Stimuli, apparatus and procedure

Stimulus presentation, event timing and response registration were controlled by the Experimental Run Time System software (ERTS, Beringer, 1995). A scanning session included four blocked-point and four randomised conditions, alternated by using an ABBABAAB design counterbalanced across subjects. Each condition included five blocks of trials, for a total of 40 blocks per experiment. The scanning session began with the presentation of a 10-s blank period, followed by the appearance of a grey fixation-cross at the centre of the projection screen for 4 s. A change in luminance (from grey to white) of the fixation-cross at the beginning of an experimental block warned the subject of the presence of information about the modality (blocked-point or randomised) of stimulus presentation throughout that block. This information was given by displaying either one or five white rectangular frames, which marked the following stimulus locations along the horizontal meridian of the visual field: 0°, and 7° and 16° of eccentricity in the left and in the right hemifield. It should be pointed out that the eccentricity differences between ERP and fMRI experiments are justified by the decision to decrease the number of conditions and to limit the duration of the latter experiment. We do not think these differences are of any importance for the sake of the present study. In the randomised condition, one location marker was displayed on each of the five field locations to indicate that the stimulus would randomly appear at one of these possible positions. In the blocked-point condition only one location marker was displayed on one of the five field positions to indicate that the stimulus would be presented only at that field location. In the blocked-point condition there was one experimental block for each field location, with the succession of positions following a fixed left-to-right or right-to-left sequence. These sequences were alternated in the four block-point conditions of an experiment by using an ABBA design counterbalanced across subjects.

The location markers were displayed for 4 s at the beginning of each block. They were extinguished thereafter, and a 4-s white fixation-cross period followed before the start of the first trial of the block. The white fixation-cross remained visible during the whole experimental block. Each block included ten event and two null-event trials randomly intermingled with the constraint of not presenting the two null-event trials consecutively. In each event trial a stimulus, consisting of a black and white checkerboard, was displayed for 100 ms to one of the possible field locations. While in the blocked-point condition the stimulus was presented to the same position across all the trials, as indicated by the location marker at the beginning of the block, in the randomised condition the stimulus location varied unpredictably on each trial with the only constraint to have two repetitions for each stimulus location. No stimuli were presented in the null-event trials. In each blocked-point and randomised condition ten stimuli were presented per field location for a total of 40 stimuli per five field locations per two conditions in one experiment. There were also 40 null-event trials per two conditions in one experiment. Each trial lasted 3 s, and the visual target was presented after an interval varying between 0, 150, 300, 450, and 600 ms from the onset of the trial. These intratrial intervals were randomly intermingled in any block with the only constraint to have two repetitions of each interval per block and two repetitions of each interval per field location per condition (for a total of eight repetitions per five intervals per five field locations per two conditions). There was an 8-s grey cross period of rest between blocks and a 20-s blank period at the end of the last block. The scanning session lasted 35 min 7 s with each block of stimuli lasting 36 s. (See Fig. 2 for the general structure of the fMRI experiment and a comparison between the paradigms used in the fMRI and in the ERP studies).

Participants were asked to keep fixation onto the central cross throughout the whole scanning session, though they were told that the targets could appear during the white cross periods only. Subjects were instructed to make a speeded button-press response with their right index finger to the appearance of a checkerboard. They were also requested to attend to the location markers appearing together with the white cross, which indicated whether in the next block of trials the checkerboard appeared always on one and the same location on the screen, as marked by one frame only, or whether it randomly appeared to one of five locations, as marked by the five frames.

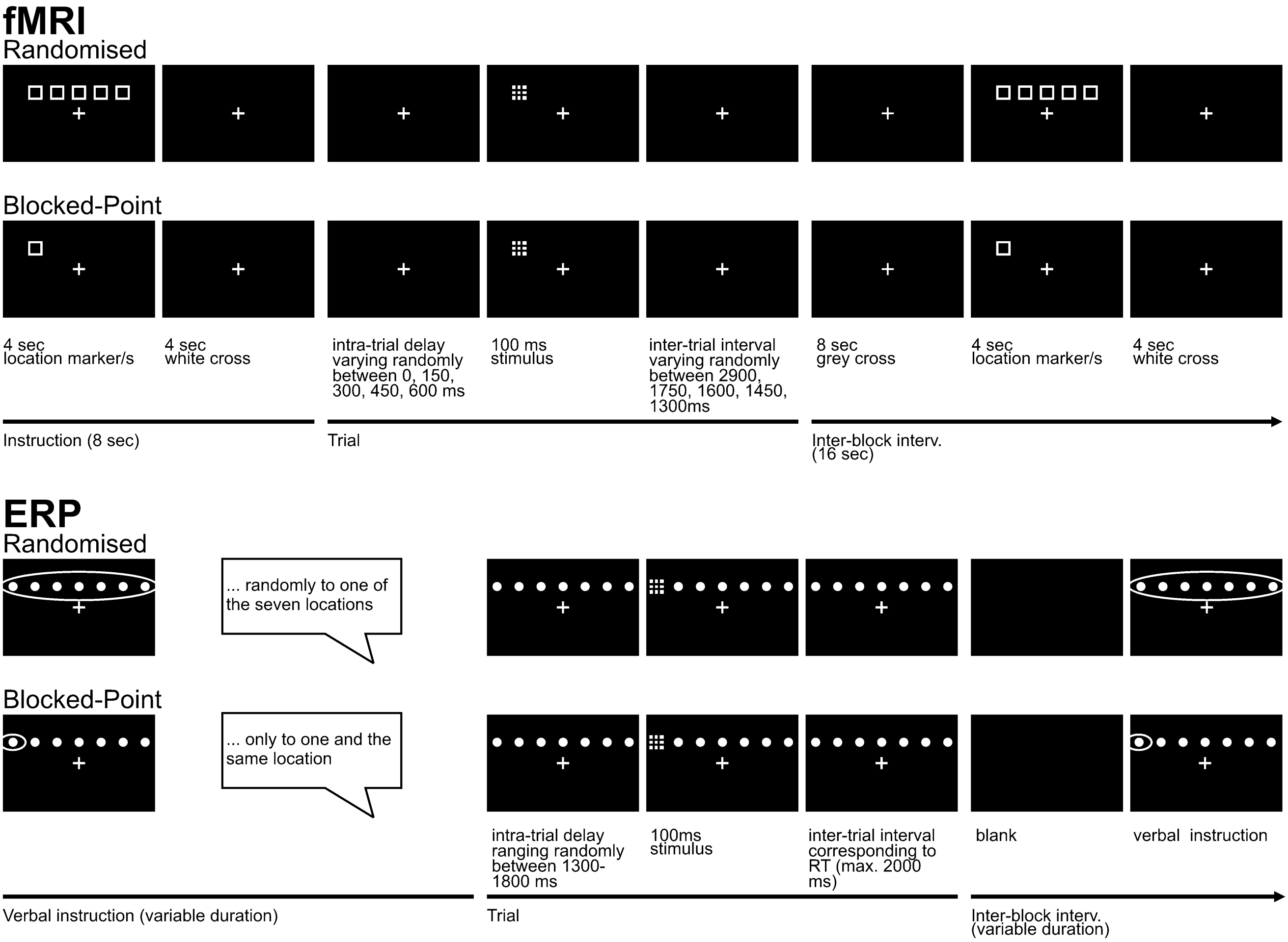

FIG. 2. Sequence of the blocked-position and randomised conditions in the fMRI and in the ERP experiment.

MRI scanning procedure

Functional images were collected at 3T with a Siemens Trio system. A gradient-echo EPI sequence was used with a TE 30 ms, a flip angle of 90° and a TR 2000 ms. The matrix acquired was 64 * 64 with a FOV of 19.2 cm, resulting in an in-plane resolution of 3 mm * 3 mm. The slice thickness was 4 mm with an interslice gap of 1 mm. Twenty-four axial slices were acquired, orientated parallel to the AC–PC plane.

To improve the effective sampling rate, the stimulus onset was varied in relation to the timing of image acquisition (Josephs *et al*., 1997; Miezin *et al*., 2000). Due to the difference in trial-length of 3 s and TR of 2 s, one-third each of stimuli was presented at the beginning of image acquisition, or 1 s or 2 s later.

Data analysis

Analysis of fMRI data was performed using the LIPSIA software package (Lohmann *et al*., 2001). Movement artefacts were corrected using a matching metric based on linear correlation. Then, a correction for slice acquisition order using sinc interpolation was performed. The data were spatially smoothed with a Gaussian kernel with FWHM 7 mm. All functional data sets were individually registered into 3D-stereotactic coordinate system using the subjects' individual high-resolution anatomical images. The 2D-anatomical slices, geometrically aligned with the functional slices, were used to compute a transformation matrix, containing rotational and translational parameters that register the anatomical slices with the 3D reference data set. Geometrical distortions of the EPI-T1 images were corrected using additional EPI-T1 refinement on the transformation matrices. These transformation matrices were normalized to the standard Talairach brain size (Talairach & Tournoux, 1988) by linear scaling, and then finally applied to the individual functional data. The normalized 3D-datasets had an isomorphic voxel size of 3 mm side length.

The statistical evaluation was based on the general linear model for serially autocorrelated observations (Friston *et al*., 1995; Worsley & Friston, 1995). For each individual subject, statistical parametric maps (SPM) were generated. The design matrix for epoch-related analysis was created using half sines convolved with a model of the haemodynamic response. The design matrix for event-related analyses was created using a synthetic model of the haemodynamic response function (Josephs & Henson, 1999), which includes an estimate of temporal autocorrelation. The data were high-pass-filtered with a cutoff frequency of 1/60 Hz. The model includes an estimate of temporal autocorrelation. The effective degrees of freedom were estimated as described by Worsley & Friston (1995).

A random effects analysis was calculated (Holmes & Friston, 1998), by computing one-sample *t*-tests of contrast-maps across subjects. The significance criterion was $P = 0.0001$, uncorrected for multiple comparisons. All reported activations were significant at the level of $P \leq 0.05$ after correction for false discovery rate (Yekutieli & Benjamini, 1999).

Due to technical problems that occurred during recordings, manual RT data were available for four out of eight subjects only. According to the same criteria as the ones used in the ERP experiment, only manual RTs ranging between 130 ms and 800 ms were included in the analyses. Overall, the amount of discarded trials, including misses, was approximately 2%.

The mean RT data were analysed by using a *t*-test for related samples comparisons with attentional Condition as within-subject factor. We also carried out a two-way ANOVA using the cumulative frequency distribution of RTs as score and attentional Condition and Percentile as within-subject factors. As for the ERP experiment, we performed separate analyses on RTs to lateral and central stimuli.

Results

ERP experiment

Reaction time

A three-way ANOVA with attentional Condition (blocked-point vs. randomised), Hemifield (Left vs. Right), and Eccentricity (4°, 8°, 12°) of stimulus presentation as within-subject factors was carried out using mean RT as score. The only significant main effect was that of Condition ($F_{1,6} = 7.357$, $P = 0.035$) with the blocked-point condition yielding faster RTs than the randomised condition, (see Fig. 3 left panel). No significant interaction was found. The main effect of Condition ($F_{1,6} = 8.1$, $P = 0.029$) was also present when analysing the cumulative frequency distribution of RTs.

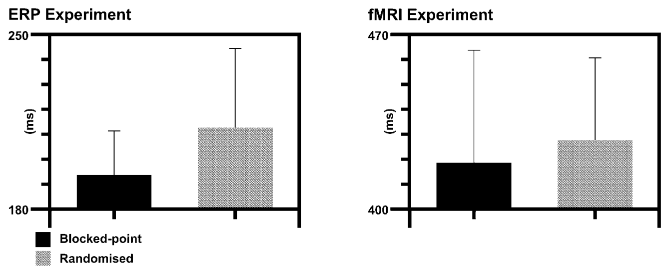

FIG. 3. Mean RT to stimuli presented in the blocked-point (black bars) and randomised (grey bars) conditions in the ERP (left panel) and fMRI (right panel) experiment. RT is averaged across the hemifield and the eccentricity of presentation.

With central stimuli, the RT difference between the blocked (185 ms) and the randomised (201 ms) condition was found to be not as reliable as with peripheral stimuli both by analysing mean RTs ($F_{1,6} = 4.2$, $P = 0.086$) and the cumulative frequency distribution of RTs ($F_{1,6} = 4.11$, $P = 0.089$).

Erp

Figure 4 shows the waveforms obtained in the blocked-point and randomised conditions. The two typical early components of the visual evoked potential, namely P1 and N1, are clearly visible. P1 is the positive-going component peaking at 110 ms from stimulus onset, and N1 is the negative-going component peaking at 170 ms from stimulus onset. The results concerning these two components will be summarized separately below.

P1. The attentional modulation of this component in the two experimental conditions was assessed by an omnibus ANOVA performed on the mean amplitude in the time-window 110–150 ms poststimulus. We found a significant interaction Condition by Hemifield by Hemisphere ($F_{1,6} = 7.39$; $P = 0.035$), indicating that the two different types of attentional condition modulated P1 as a function of the stimulated hemifield and of the hemisphere of the recording site. As can be seen in Fig. 4, only the hemisphere contralateral to the stimulated hemifield (left panel) showed the P1 attentional effect, whereas the ipsilateral hemisphere (right panel) did not. A *post hoc t*-test carried out to substantiate this observation showed a significant attentional effect, with the blocked-point condition yielding a larger amplitude than the randomised condition on the hemisphere contralateral to the stimulated visual field only ($t_6 = 2.69$; $P = 0.036$). In contrast, the two attentional conditions had no effect on the P1 recorded over the hemisphere ipsilateral to the stimulated visual field ($t_6 = 0.50$; $P = 0.6$). This attentional effect on P1 did not interact with either Eccentricity or Electrode ($P = 0.56$ and $P = 0.92$, respectively).

N1. A clear inversion of the attentional effect described above for P1 is visible in Fig. 4 (left panel) in the N1 time-window (155–195 ms poststimulus) where this component showed a larger amplitude in the randomised than the blocked-point condition. An omnibus ANOVA confirmed the difference between the two attentional conditions by showing a significant interaction Condition by Hemifield by Hemisphere [$F_{1,6} = 16.07$; $P = 0.007$]. As for the P1 component, the

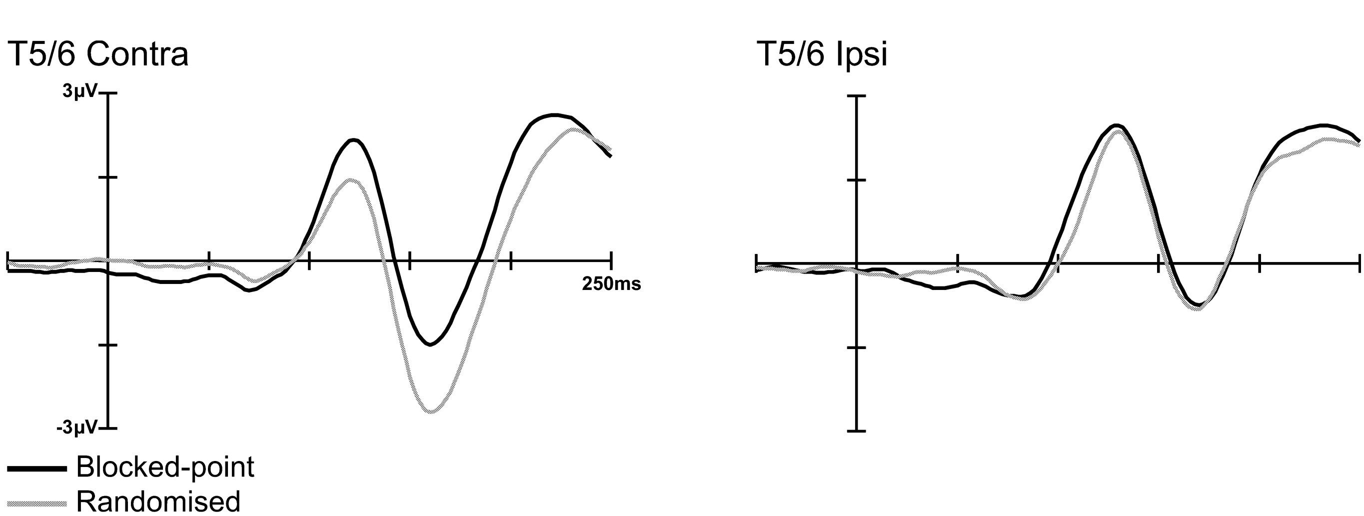

FIG. 4. ERPs waveforms elicited by blocked-point (black lines) and randomised (grey lines) stimuli, obtained from temporal sites (T5/T6) in the hemisphere contralateral (left panel) and ipsilateral (right panel) to the stimulated hemifield. ERPs were averaged across the hemifield and the eccentricity of stimulus presentation.

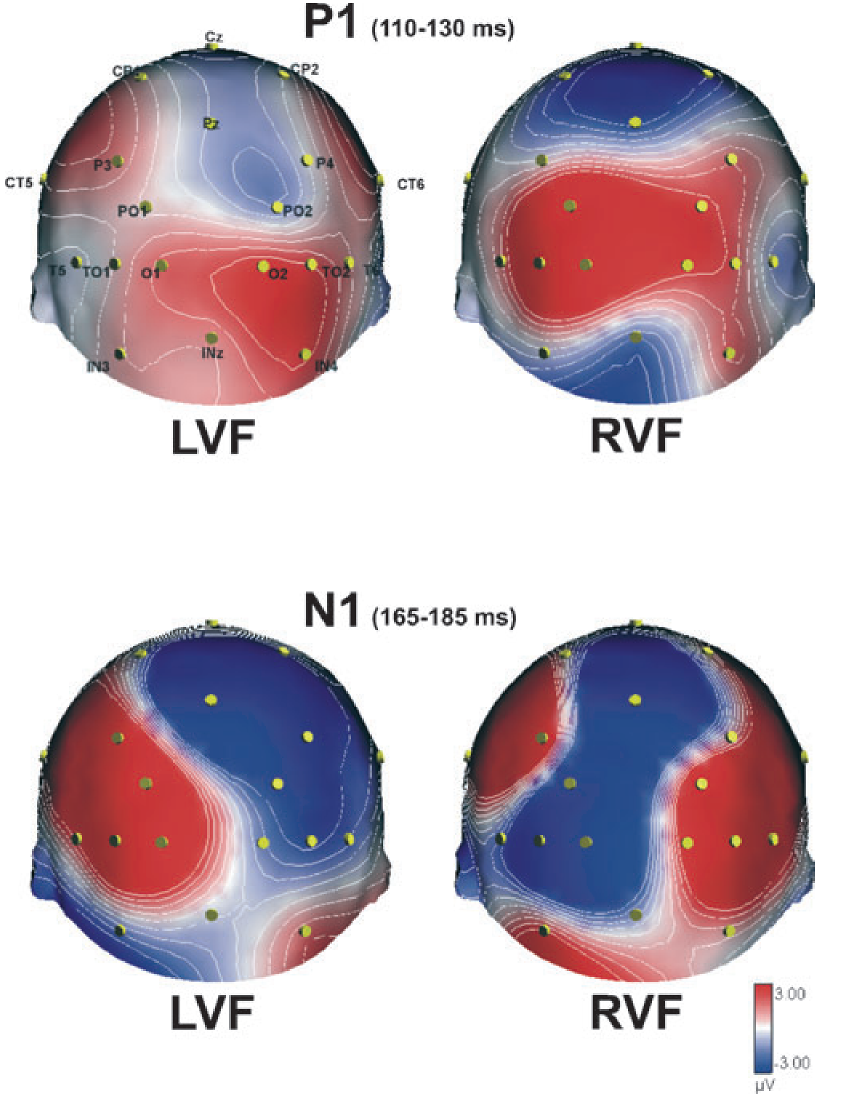

FIG. 5. CSD maps of the difference waveforms (blocked-point minus randomised) for the P1 component (top panel) and the N1 component (bottom panel) separated for attentional orienting towards left visual field (LVF) and right visual field (RVF). The maps were calculated around the peak of the component and within the time window used to measure the mean amplitude and to perform the statistical analysis.

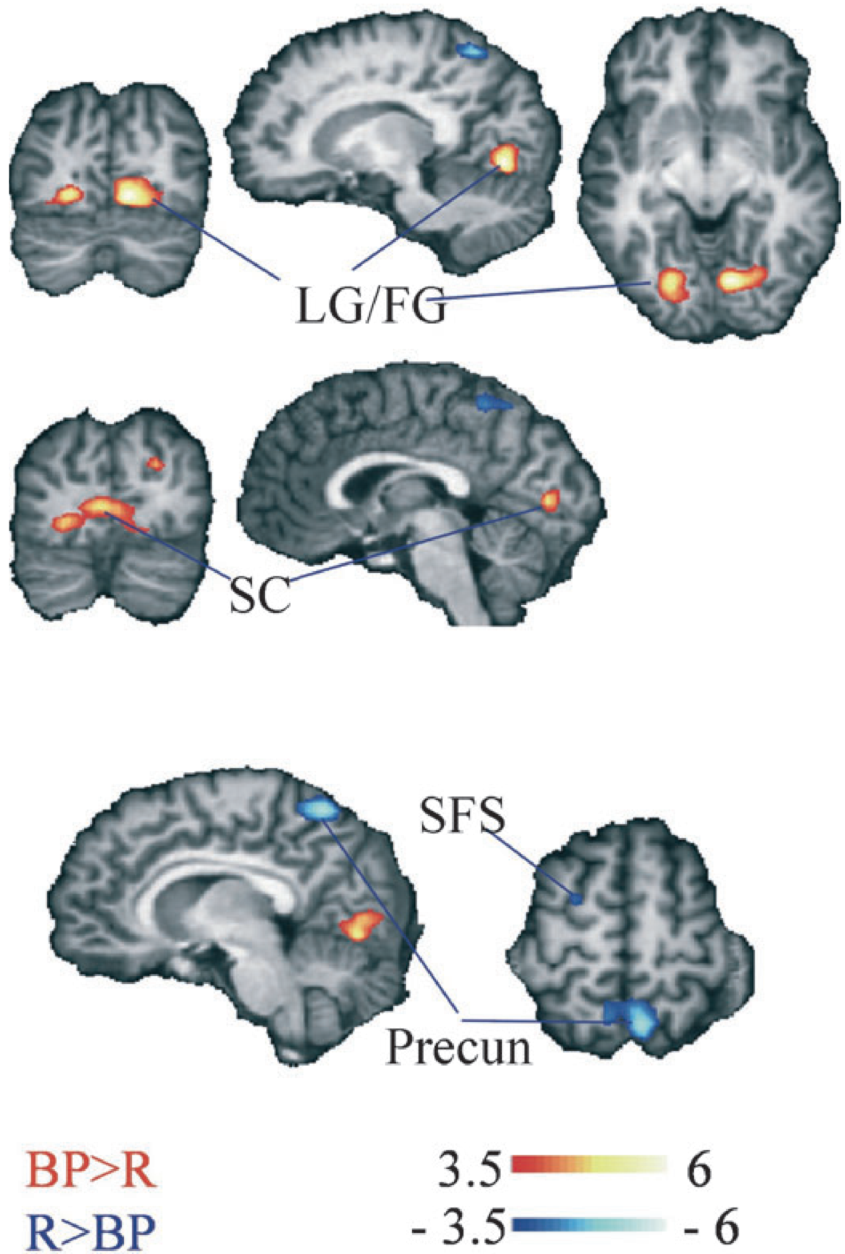

FIG. 6. fMRI activations for the contrast blocked point – randomised condition. The red scale indicates stronger activation in the blocked point condition, the blue scale stronger activation in the randomised condition. Left hemisphere is on the left in the axial and coronal slices. The legend indicates z-scores. FG, fusiform gyrus; LG, lingual gyrus; precun, precuneus; SC, striate cortex; SFS, superior frontal sulcus; BP, blocked-point condition; R, randomised condition.

presence of an attentional effect exclusively over the hemisphere contralateral to the stimulated hemifield (left panel) was confirmed in a *post hoc* t-test ($t_6 = 2.3$; $P = 0.056$). In keeping with what was found for P1, over the hemisphere ipsilateral to the stimulated hemifield (right panel) attentional condition did not exert any effect on the N1 component ($t_6 = -0.02$; $P = 0.98$).

ERP to central stimuli. Two three-way ANOVAs, one on P1 and another on N1 mean amplitude were carried out for responses to central stimuli with attentional Condition, Hemifield, and Electrode as factors. None of these analyses revealed a main effect of Condition or interactions of Condition with other factors.

Scalp topographies. The CSD maps of the attentional modulation (difference waveforms) for the two components considered, showed a clearly separated pattern of scalp distribution for P1 and N1 (see Fig. 5 top panel and bottom panel, respectively). In particular, while the P1 effect is confined to the lower occipital lateral sites, the N1 effect is much more dorsal and medial on both hemispheres. This spatial distribution is compatible with independent attentional effects elicited by the two attentional conditions, namely voluntary focusing vs. automatic orienting, on the P1 and N1 components, respectively.

fMRI experiment

Reaction time

Due to technical problems during recording, RTs were available only for four out of the eight participants in the fMRI experiment. As can be seen in Fig. 3 (right panel), RTs were overall faster with blocked than randomised presentations. However, because of the small sample size, this difference did not reach statistical significance. Again, there was no substantial difference for RTs to central stimuli in the blocked (415 ms) as compared to the randomised (419 ms) condition.

Functional imaging

In the comparison of the blocked-point minus the randomised condition, foci of statistically significant activation were observed bilaterally in the lingual gyri spreading into the fusiform gyri. In the left hemisphere the activation reached up into the lower bank of the calcarine sulcus. A further activation was observed at the junction of the right intraparietal sulcus with the transverse occipital sulcus (see Fig. 6 and Table 1).

In the reverse contrast, randomised minus blocked-point condition, a significant activation was found along the posterior portion of the left superior frontal sulcus, just anterior to the junction with the superior precentral sulcus. A second activation was observed bilaterally in the superior precuneus (see Fig. 6 and Table 1).

We compared the blockwise activation in the blocked point and the random condition against the fixation periods immediately before and between blocks (white fixation-cross periods following location markers displays, see Fig. 2). In the random condition, a large activation was observed in the left ventral occipital cortex, centred on the fusiform gyrus. Additional activations were observed in the lateral occipital gyrus and precuneus, both in the left hemisphere (see Table 1). Comparison of the blocked point condition to fixation yielded no significant activation after correction for false discovery rate. There were no significant signal decreases.

Likewise, comparing the blocked point epochs for the two leftmost with the two rightmost positions did not yield significant activation. The same was true for the event-related comparison of the two leftmost and rightmost positions in the random condition.

TABLE 1. fMRI activations for the blocked-point – randomised, randomised – fixation, randomised left – randomised right contrasts

Contrasts and structures	Hemisphere	Coordinates x	y	z	Zmax
Blocked-point – randomised					
Fusiform gyrus	L	−17	−76	−3	5.22
IPS/TOS	R	25	−82	24	4.45
Lingual gyrus	R	13	−76	−6	6.01
Striate cortex	L	−2	−82	5	4.76
Randomised – blocked-point					
Superior frontal sulcus	L	−23	2	56	4.14
Precuneus	R	8	−61	53	5.38
Precuneus	L	−5	−53	51	4.28
Randomised – fixation					
Precuneus	L	−11	−76	30	3.95
Fusiform gyrus	L	−17	−70	0	4.57
Lateral occipital gyrus	L	−32	−76	−3	3.90
Randomised, Left – Right					
Superior frontal gyrus (medial aspect)	R	−2	−1	57	3.65
Intraparietal sulcus, descending segment	R	25	−85	36	4.41
Superior frontal sulcus	L	−32	23	45	−3.66
Superior parietal lobule	L	−11	−70	54	−3.31
Intraparietal sulcus, descending segment	L	−8	−91	36	−3.62
Posterior cingulate cortex	L	−17	−40	30	−3.48
Posterior cingulate cortex	L	−5	−52	21	−3.44
Lateral occipital gyrus	L	−38	−79	18	−3.87
Fusiform gyrus	L	−29	−58	0	−4.77
Fusiform gyrus	L	−23	−70	0	−3.12
Precuneus	R	−2	−64	42	−3.98
Inferior parietal lobule	R	34	−70	39	−3.95
Posterior cingulate cortex	R	4	−37	36	−3.29

L, left-hemisphere; R, right-hemisphere; IPS/TOS, intraparietal sulcus/transverse occipital sulcus; Zmax, local maximum of Z-score.

In order to search for tendencies of lateralization in known attention-related areas, we lowered our threshold to $P > 0.0001$ for a restricted search in areas typically activated by visuospatial attention. We observed stronger activation for left compared to right hemifield positions along the midline at the border between preSMA and SMA and along the descending segment of the right intraparietal sulcus. A reverse, stronger activation elicited by right hemifield trials, was observed in the left hemisphere along the superior frontal sulcus, the descending segment of the intraparietal sulcus, superior parietal lobule, posterior cingulate cortex, lateral occipital and fusiform gyri. There was also right hemispheric activation in the precuneus, inferior parietal lobule and posterior cingulate cortex (see Table 1).

Discussion

In the present study, we employed a speeded detection task and compared RT, ERP and fMRI responses to the same stimulus presented at predictable (blocked) vs. unpredictable (randomised) field locations. The manipulation of the predictability of stimulus position was intended to modify the subject's attentional set so that in the blocked-point condition attention would be sustained onto one stimulus position, whereas in the randomised condition it would be diffused across the whole array prior to stimulus presentation and subsequently drawn by stimulus appearance.

We obtained RT, ERP, and fMRI evidence supporting the functional distinction between endogenous, sustained focusing of attention, and exogenous, transient allocation of spatial attention. As predicted by the idea of a differential attention allocation in the two conditions, we found an overall RT advantage of the blocked-point over the randomised condition, thus replicating our previous findings obtained with the same paradigm (Marzi *et al*., 2002; Natale *et al*., 2005). However, this was true only with peripheral stimuli, whereas with central stimuli the two conditions did not yield a difference in RT as reliable as with peripheral stimuli. The latter finding is not surprising, given that central stimuli are very close to fixation, hence, into the focus of attention under both conditions. This is because even in the randomised condition, participants, while diffusing their attention across the whole array, might keep most of their attentional resources by default onto the central location. As far as peripheral stimuli are concerned, the RT facilitation under condition of sustained attention was associated with an increase in the amplitude of the posterior P1 component of ERP. This effect is in agreement with other studies of sustained attention (see Mangun *et al*., 1993; Hillyard & Anllo-Vento, 1998 for reviews) showing that spatial attention modulates the visual input at an early level of perceptual processing, rather than at a postperceptual stage (Mangun, 1995). On the contrary, a larger amplitude for the randomised rather than the blocked-point condition was observed for the posterior N1 component. In keeping with the absence of behavioural effects, no modulation of P1 and N1 with attentional condition was observed for central stimuli.

The differential modulation of P1 and N1 that was observed over the hemisphere contralateral to the visual hemifield of stimulus presentation strongly supports the possibility that the posterior P1 and N1 components of ERP reflect different attentional mechanisms. While the P1 effect might index a facilitation of early sensory processing for stimuli presented at the location where attention is endogenously focused, the N1 effect might represent a transient allocation of attention captured by the appearance of the stimulus. This hypothesis has already been put forward to explain the evidence of a P1 and N1 dissociation found in previous ERP studies (Hillyard *et al*., 1985; Heinze *et al*., 1990; Luck *et al*., 1990; Mangun & Hillyard, 1991; Heinze & Mangun, 1995). Importantly, part of this evidence could be explained by different accounts, which, on the contrary, do not seem to fit with our results. In particular, it has been found that the N1 effect is elicited by stimuli at the to-be-attended location preceded by isolated events appearing at the unattended side during sequences of unilateral presentations randomly delivered to either side, whereas it is absent or reduced when all-bilateral or mixed (bilateral and unilateral) sequences of stimuli, respectively, are used (Heinze *et al*., 1990; Luck *et al*., 1990). This has been interpreted as evidence that N1 indexes attentional shifts, as attention has to be switched back to the to-be-attended location after having been captured by an event appearing at the unattended side in an otherwise empty field. As an alternative, it has been argued that the reduction of the N1 effect observed with bilateral presentations might simply be a consequence of having increased the repetition rate of visual stimuli at the to-be-attended position and represent either a limited capacity effect or a result of sensory refractoriness due to stimulus repetition (Heinze *et al*., 1990; Luck *et al*., 1990). A similar argument could apply to the reduction of the amplitude of N1 observed in the blocked-point condition of the present study where the repetition rate at each spatial location is clearly higher than in the randomised condition. However, one should consider that, in contrast to previous studies (Luck *et al*., 1990) which used very short interstimulus intervals (ranging from 310 to 450 ms), we have chosen for both attentional conditions rather long interstimulus intervals (ranging randomly between 1300 and 1800 ms plus the RT of the subject). This was done just with the intent of minimizing ERP modulations due to sensory refractoriness, limited capacity processing or any overlap with ERP responses to preceding stimuli (Woldorff, 1993). Furthermore, in the study of Luck *et al*. (1990) a behavioural counterpart of a limited capacity process was documented, in that target detection was slower and less accurate under those conditions where the N1 effect was reduced. In contrast, we found the opposite pattern of correlation between the behavioural and electrophysiological modulations, with faster RTs under the blocked-point condition, where the N1 amplitude was found to be reduced, and slower RTs under the randomised condition, where we found an enhanced negativity. Finally, it is unclear why in our study as well as in that of Luck *et al*. (1990) a refractory effect should selectively eliminate the modulation of sustained attention on N1 while preserving that on P1.

According to another account, a broad contralateral positivity lasting longer than the P1 latency range might be associated with sustained attention. As an effect of such a positivity, which may represent either a modulation of the exogenous P1 wave overlapping with a later positivity or a unitary endogenous wave, any other electrophysiological effect yielding a negative component and occurring in the same latency range would be cancelled. This would explain the absence of the N1 effect under the blocked-point condition of the present experiment as well as the ‘apparent’ presence of the N1 effect under the randomised condition where ERP waves might simply be on average less positive than in the blocked condition. Heinze & Mangun (1995) have shown that when over contralateral scalp regions the N1 effect is cancelled by an overlapping prolonged positivity, it may be still evident over the ipsilateral scalp. However, this was not the case in our experiment where at ipsilateral sites we did not observe either a larger negativity in the blocked-point than in randomised condition or an overall larger negativity than at contralateral sites. Furthermore, there is evidence (Lange *et al*., 1999) indicating that the prolonged contralateral positivity also found in studies with bilateral stimuli (Heinze *et al*., 1990; Luck *et al*., 1990) might be a consequence of lateralized activity related to a search process among the multiple stimuli typically displayed with bilateral arrays. As in the present study subjects were not required to perform either any display search or any high-order discrimination, the different modulation of the amplitude of P1 and N1 found with randomised and blocked presentations is likely to represent the effect of spatially specific mechanisms of attention selection.

In principle, it remains possible that the overall difference in amplitude between ERP waveforms in our two conditions might be due to eccentric fixation favoured by prior knowledge of stimulus location in the blocked-point condition. However, as we have on-line monitored subjects’ fixation and rejected trials with saccades triggered by stimulus appearance, we can confidently exclude that the ERP effects may be attributed to systematic or stimulus driven shifts of fixation. As illustrated in Fig. 7, overall, the HEOG and VEOG waveforms are aligned with the zero line in the two conditions at and beyond the latencies of the P1 and N1 components. Tiny deviations from the zero line, visible at the end of the time window, are in the order of one- or two-tenths of degree. Moreover, the HEOG and VEOG waveforms perfectly overlap in the two conditions at and beyond the latencies of the P1 and N1 components, indicating that in the two attentional conditions the subjects did not behave differently and maintained fixation steadily throughout.

In the fMRI experiment, randomised and blocked-point conditions yielded selective activation in different brain areas. Attention shifts to different locations in the randomised condition were accompanied by signal increase in more anterior areas, such as frontal and posterior parietal areas, with respect to the blocked-point condition. In the latter

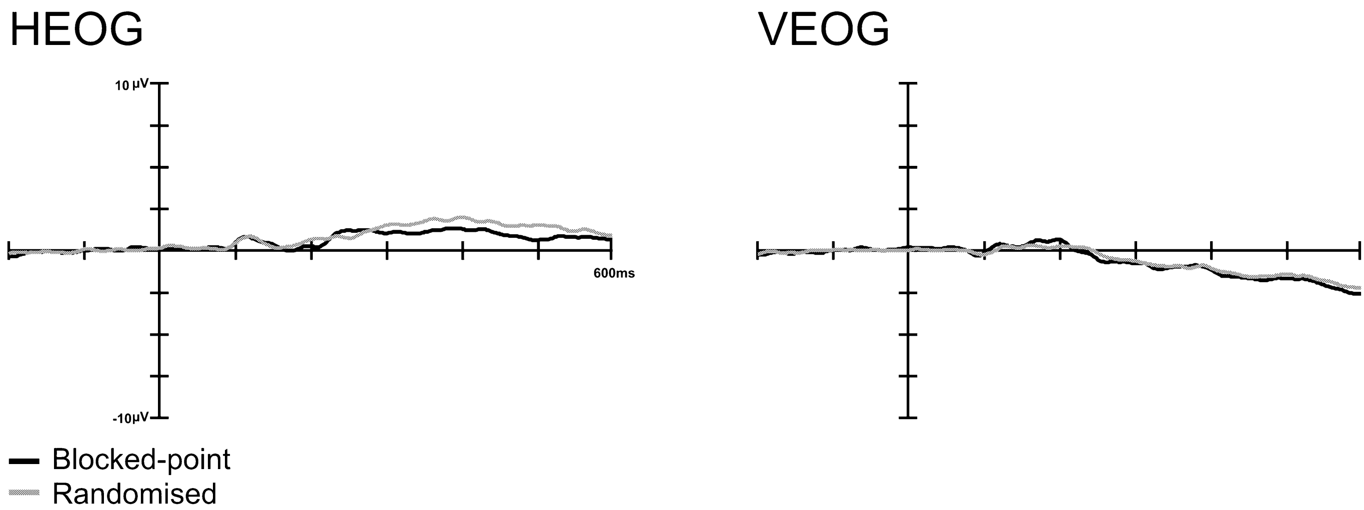

FIG. 7. HEOG and VEOG waveforms for the Blocked-point (black) and Randomised (grey) condition, averaged across hemifield and eccentricity of stimulus presentation.

the activation was confined to occipital areas, reaching down as far as striate cortex. Contrasts of the experimental conditions against fixation assured that the observed activations were not due to signal decreases in these areas. The absence of significant signal increases in the blocked point condition compared to the baseline may be due to preparatory attention occurring during the fixation baseline, which followed instruction (location markers) as to the condition of presentation (see Fig. 2). Lateralized activations elicited by contralateral stimulus presentation in areas belonging to the network subserving visuospatial attention were observed (though not statistically significant for the whole head analysis) in the random condition, in keeping with the possibility that the random spatial stimulus onsets triggered an automatic deployment of attention.

These results confirm the concept that the frontal and parietal parts of the attention network are involved in eliciting attention changes, whereas the effects of these attention changes are seen as a modulation of activation strength in visual occipital areas (Kastner *et al*., 1999). The association between increased activity in occipital areas and sensory facilitation related to sustained attention with blocked presentations is in keeping with fMRI (Mangun *et al*., 1998; Martínez *et al*., 1999; Martínez *et al*., 2001; Noesselt *et al*., 2002) and PET (Heinze *et al*., 1994; Mangun *et al*., 1997; Woldorff *et al*., 1997; Mangun *et al*., 2001) studies in humans as well as with electrophysiological recordings in animals demonstrating that spatial attention can influence neuronal firing rates in various extrastriate areas (see Desimone, 1998; Maunsell & McAdams, 2000 for reviews). Interestingly, several studies combining ERP with fMRI or PET found an attention-related contralateral enhancement of posterior P1 but not the N1 component (Heinze *et al*., 1994; Mangun *et al*., 1997; Woldorff *et al*., 1997; Mangun *et al*., 1998; Noesselt *et al*., 2002), suggesting that in tasks of sustained attention there is a relationship between the P1 effect and the neural activity changes in extrastriate cortical areas. Importantly, the large activation that we found in lingual and fusiform gyrus is in accordance with evidence indicating this area as a possible generator of P1, namely, of a component that is modulated by selective attention (Heinze *et al*., 1994). Likewise, the cortex at the banks of the intraoccipital sulcus, where we found increased activation in the blocked-point condition, has also been considered as another possible generator of P1 (Martínez *et al*., 1999). Thus, our data indicate that the sustained attention-related effects observed in our two experiments, namely the enhanced amplitude of the posterior P1 component of the ERP and the increased fMRI activity in occipital areas, are mutually associated, confirming evidence from studies directly combining temporal and spatial imaging of brain activity during sustained attention (Heinze *et al*., 1994; Mangun *et al*., 1997; Woldorff *et al*., 1997; Mangun *et al*., 1998; Noesselt *et al*., 2002) and corroborating evidence showing that generators of the posterior P1 are located in both ventral and dorsal regions of occipital cortex. Less is known about the source localization of N1, and we can only speculate that the bilateral precuneus and the cortex at the banks of the left superior frontal sulcus, which were differentially activated in the randomised condition, may be involved in generating the N1 effect observed in the present study.

In conclusion, we believe that our results represent reliable evidence of a functional dissociation between attentional effects on posterior P1 and N1 components of ERP, with the former indexing an effect of endogenous-sustained attention and the latter an effect of an exogenous-transient orienting of attention. A different account of P1–N1 functional dissociation has been proposed indicating that P1 and N1 attention effects reflect suppression and enhancement of sensory processing, respectively (Luck, 1995). Interestingly, the P1 suppression effect has been specifically reported either with a spatial cueing or a visual search paradigm under conditions where unattended stimuli were present, which could provide a substantial source of confounding noise (see Luck, 1995 for a more detailed discussion about this issue). Therefore, it is reasonable to conclude that P1 attention effects reflect sensory processing facilitation, which can operate either by means of a preset bias and sustained gating of input at the attended location or by mechanisms of noise suppression at the unattended location according to contingent factors, like available information and task demand. On the other hand, the functional significance of the attention-related N1 modulation seems to be less clear than that of P1. It has also been suggested that N1 would index a discrimination process within the focus of attention (Luck, 1995; Vogel & Luck, 2000). Here, we propose that the N1 effect could reflect mechanisms directly involved in orienting of attention rather than in its consequences. In particular, we suggest the possibility that the posterior N1 indexes an exogenous, stimulus driven orienting of attention. This hypothesis is supported by evidence from previous ERP studies, see above, and from a study of Yamaguchi *et al*. (1995) wherein a comparison between stimuli presented at predictable vs. unpredictable locations showed no significant difference in P1 amplitude for all scalp sites and an enhancement of N1 over contralateral temporal sites specifically elicited by stimuli presented to unexpected locations. Similarly, Fu *et al*. (2005) by using peripheral cues to direct participants' attention involuntarily to a spatial location during a line-orientation discrimination task have found that invalid trials elicited larger posterior contralateral N1 relative to valid trials, possibly representing a stimulus-driven shift of attention from the valid location. A larger contralateral P1 was elicited by valid compared to invalid trials and might reflect suppression of irrelevant information while searching for a target among distracter stimuli within the same spatial location. Further evidence for our above hypothesis comes from studies with visual extinction and/or neglect patients, where a smaller N1 component was elicited by contralesional than ipsilesional target stimuli (Marzi *et al*., 2001) or cues (Verleger *et al*., 1996) as well as by the combination of right cue/left target as compared to other combinations of cue and target (Verleger *et al*., 1996). In our study (Marzi *et al*., 2001), this was associated with a dramatic interfield asymmetry of visual detection, with slower and less accurate responses to contra- than ipsilesional stimuli. Several lines of behavioural evidence indicate an impairment of exogenous attention as responsible for the interfield asymmetry of visuo-motor reactions in neglect patients (Natale *et al*., 2005), whose electrophysiological correlate may be represented by the reduction of the N1 amplitude.

Several neuroimaging studies comparing attention shifts triggered by the presentation of peripheral vs. central attention-directing cues have provided evidence that a large-scale neural network (including prefrontal, posterior-parietal, superior temporal and cingulate cortices as well as the insula) subserves both top-down and bottom-up mechanisms of attention control (Corbetta *et al*., 1993; Nobre *et al*., 1997; Kim *et al*., 1999; Rosen *et al*., 1999; Peelen *et al*., 2004). In contrast, recent fMRI (Mayer *et al*., 2004) as well as psychological, neuropsychological and physiological studies (Corbetta & Shulman, 2002) have suggested that separate neural systems control endogenous and exogenous modes of orienting. In keeping with that, Yamaguchi *et al*. (1994) found that both central and peripheral cues generated a sustained, late negative potential shift in the hemisphere contralateral to the direction of attention, whose scalp distribution, onset, and duration differed with the type of cue. Moreover, an earlier negative potential shift, namely a contralateral increase of N1 (140–200 ms) over both anterior and posterior scalp sites, was observed only with peripheral cues, indicating a possible association of N1 with an exogenous shift of spatial attention. However, whether the posterior N1 would specifically index an exogenous shift of attention or whether it may represent the activity of a large scale attentional neural network possibly subserving both exogenous and endogenous orienting of attention, remains to be ascertained.

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Abbreviations

CSD, current source density; ERP, event-related potential; fMRI, functional magnetic resonance imaging; HEOG, horizontal electrooculogram; RT, reaction time; VEOG, vertical electrooculogram.

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